Name: ____________________

VOLUME AND SURFACE AREA, (VOLUME OF PRISMS AND CYLINDERS)

Date: ________________

LO: I can calculate the volume of prisms and cylinders.

Volume of prisms

The **volume** of a prism = **cross-section area × length**. For a cylinder, $V = \pi r^2 h$.

Units: Volume is measured in cubic units (cm^3, m^3).

Key formulae

●	Prism: V = area × length	cross-section × depth
●	Cylinder: $V = \pi r^2 h$	circle area × height
●	Cuboid: V = lwh	special case of prism
●	$V = 2\pi rh$	that is curved surface area
●	Cylinder $V = \pi rh$	must square the radius

Key idea

Volume = cross-section area × length (or height for vertical prisms).

$$V = A \times l$$

A = area of cross-section, l = length/height

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – cuboid

Find the volume of a cuboid 8 × 5 × 3 cm.

$V = l \times w \times h$

$= 8 \times 5 \times 3$

$= 120\ cm^3$

$\boxed{120\ cm^3}$

Multiply length × width × height.

Example 2 – cylinder

Find the volume of a cylinder r=4, h=10.

$V = \pi r^2 h$

$= \pi \times 16 \times 10$

$= 160\pi = 502.7\ cm^3$

$\boxed{502.7\ cm^3}$

Square the radius, then multiply by π and height.

Example 3 – triangular prism

Triangle: base 6, height 4. Prism length 10. Volume?

Cross-section = $\frac{1}{2} \times 6 \times 4 = 12\ cm^2$

$V = 12 \times 10$

$= 120\ cm^3$

$\boxed{120\ cm^3}$

Find the triangle area first, then multiply by length.

Example 4 – finding height

Cylinder V=300π, r=5. Find h.

$300\pi = \pi \times 25 \times h$

$300 = 25h$

$h = 12\ cm$

$\boxed{h = 12\ cm}$

Rearrange: $h = V/(\pi r^2)$.

YOUR TURN – PRACTICE (deliberate practice)

1. Cuboids
Find the volume.

a) $5 \times 4 \times 3$

b) $10 \times 6 \times 2$

c) $8 \times 8 \times 8$

d) $12 \times 5 \times 4$

2. Cylinders
Find the volume (1 dp).

a) r=3, h=7

b) r=5, h=12

c) r=2, h=15

d) d=10, h=8

3. Triangular prisms
Find the volume.

a) Triangle: b=8, h=3, length=10

b) Triangle: b=6, h=5, length=12

c) Triangle: b=10, h=4, length=7

d) Right triangle: 3,4,5, length=9

4. Finding missing dimensions
Find the unknown.

a) Cuboid V=180, l=9, w=5. Find h.

b) Cylinder V=200π, r=5. Find h.

c) Prism V=240, cross-section=30. Length?

d) Cube V=343. Find side.

5. Mixed
Solve each.

a) Swimming pool 25×10×2 m. Volume in litres?

b) Cylindrical tank r=3m, h=5m. Volume?

c) Prism: trapezium (a=4,b=8,h=3), length=10

d) Cylinder V=500 cm^3, h=10. Find r.

e) Cube SA=150. Find volume.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Volume of a prism = cross-section area × length ☐	b) Cylinder volume = πrh ☐	c) Volume is measured in cubic units ☐	d) Doubling the radius doubles the volume ☐	e) A cuboid is a type of prism ☐
Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________
Reason: __________________	Reason: __________________	Reason: __________________	Reason: __________________	Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A cylindrical glass has radius **4 cm** and height **12 cm**. It is filled to $\frac{3}{4}$ full. How many ml of water does it contain? (1 cm^3 = 1 ml)

Expression: ______________________________

Simplified: ______________________________

b) A prism has an L-shaped cross-section made from two rectangles: 6×2 and 4×2 (total area 20 cm^2). The prism is 15 cm long. Find its volume and total surface area.

Expression: ______________________________

Simplified: ______________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!